Causal Inference
Lecture 6

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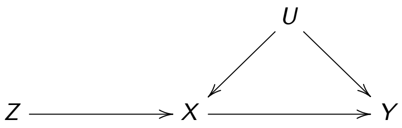
Identification and Estimation of the causal effect $X \to Y$

The examined problem:

$$\begin{tikzcd} & U \arrow[dl] \arrow[dr] & \\ X \arrow[rr] & & Y \end{tikzcd}$$

- Simple causal structure where U confounds the effect of X on Y.
- Our goal is to identify and estimate the causal effect of X on Y.
- We assume that U is unmeasured (unobserved).

- When U is unmeasured, one can still identify the causal effect of interest by using a special variable Z.
- Such a variable is associated monotonically with the exposure X (relevance), affects the outcome Y only through the exposure (exclusion), and is not confounded with the outcome (exogeneity).
- These three properties define a valid instrumental variable (IV). Such variables allow for the identification of causal effects in the presence of unmeasured confounders U.



DAG of a valid instrumental variable Z

A linear causal model

- The structural model is
$$Y = \psi_0 + \psi_1 X + f(U) + u$$
where the parameter of interest is ψ_1.
- This is also the causal model, i.e.,
$$Y_x = \psi_0 + \psi_1 x + f(U) + u$$
where ψ_1 is the average treatment effect (ATE)
$$\mathbb{E}[Y_{x+1} - Y_x] = \mathbb{E}[\psi_1(x+1) + f(U) + u - \psi_1 x - f(U) - u] = \psi_1.$$
- Namely, ψ_1 is the target parameter of the analysis.

Ordinary least squares (OLS) derivation

- Let $y_i = \beta_0 + \beta_1 + \epsilon$, $i = 1, ..., n$, be the assumed model for the observed data $\{(y_1, x_i)\}_{i=1}^n$.
- The classical model assumptions are $\epsilon_i | X \sim^{iid} N(0, \sigma^2)$. Namely, the error terms are independent identically distribution normal random variables (conditionally on X).
- The OLS are the values of β that minimize the sum of the squared errors (or the MSE), i.e.,

$$\arg\min_{\beta \in \mathcal{B}} \sum (y_i - \beta_0 - \beta_1 x_i)^2 = \arg\min_{\beta \in \mathcal{B}} S(\beta)$$

- By taking derivative w.r.t. the vector β we obtain the normal equations

$$\frac{\partial}{\partial \beta_0} S(\beta) \propto \sum (y_i - \beta_0 - \beta_1 x_i) = 0,$$
$$\frac{\partial}{\partial \beta_1} S(\beta) \propto \sum (y_i - \beta_0 - \beta_1 x_i) x_i = 0.$$

the values of β_0 and β_1 that solve these equations are the OLS estimators of β_0 and β_1, respectively.

Ordinary least squares (OLS) derivation

- By solving the normal equations we get the following results

$$
\begin{aligned}
\hat{\beta}_1 =& \frac{\sum(x_i - \bar{x}_n)(y_i - \bar{y}_n)}{\sum(x_i - \bar{x}_n)^2} = \frac{\widehat{cov}(X, Y)}{\widehat{var}(X)} \\
\hat{\beta}_0 =& \bar{y}_n - \hat{\beta}_1 \bar{x}_n
\end{aligned}
$$

  which are the OLS estimators of β_1 and β_0, respectively.

- Alternative form of the OLS of β_1

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\sum(x_i - \bar{x})(\beta_0 + \beta_1 x_i + \epsilon_i)}{\sum(x_i - \bar{x})^2} \\
&= \beta_0 \frac{\sum(x_i - \bar{x})}{\sum(x_i - \bar{x})^2} + \beta_1 \frac{\sum(x_i - \bar{x})^2}{\sum(x_i - \bar{x})^2} + \frac{\sum(x_i - \bar{x})\epsilon_i}{\sum(x_i - \bar{x})^2} \\
&= \beta_1 + \sum w_i \epsilon_i,
\end{aligned}
$$

  namely, the OLS equals the true β_1 plus weighted sum of normal random variables.

Definition: Unbiasedness An estimator $\hat{\beta}_j$ (for $j = 0, 1$) is said to be unbiased if:

$$E[\hat{\beta}_j] = \beta_j$$

where β_0 and β_1 are the true population parameters.
Proof: for $\hat{\beta}_1$,

$$\begin{aligned}
\mathbb{E}[\hat{\beta}_1|X] &= \mathbb{E}[\frac{\sum_i (x_i - \bar{x}) y_i}{\sum_i (x_i - \bar{x})^2}|X] \\
&= \frac{\sum_i (x_i - \bar{x})(\mathbb{E}[\beta_0 + \beta_1 x_i + \epsilon_i|X])}{\sum_i (x_i - \bar{x})^2} \\
&= \frac{\sum_i (x_i - \bar{x})(\beta_0 + \beta_1 x_i)}{\sum_i (x_i - \bar{x})^2} \\
&= \beta_0 \frac{\bar{x}_n - \bar{x}_n}{\sum (x_i - \bar{x})^2} + \beta_1 \frac{\sum (x_i - \bar{x})^2}{\sum (x_i - \bar{x})^2} = \beta_1.
\end{aligned}$$

Alternatively

$$\mathbb{E}[\hat{\beta}_1|X] = \mathbb{E}[\beta_1 + \sum w_i \epsilon_i|X] = \beta_1 + \sum w_i \mathbb{E}[\epsilon_i|X] = \beta_1.$$

Proof: for $\hat{\beta}_0$,

$$
\begin{aligned}
\mathbb{E}[\hat{\beta}_0|X] &= \mathbb{E}[\bar{y} - \hat{\beta}_1\bar{x}|X] \\
&= \mathbb{E}[\frac{1}{n}\sum(\beta_0 + \beta_1 x_i + \epsilon_i) - \beta_1\bar{x}_n|X] \\
&= \mathbb{E}[\beta_0 + \beta_1\bar{x}_n + \bar{\epsilon} - \beta_1\bar{x}_n|X] \\
&= \beta_0.
\end{aligned}
$$

The variance of $\hat{\beta}_1$ is:

$$\begin{aligned}
\mathrm{Var}(\hat{\beta}_1|X) &= \mathrm{Var}\left(\frac{\sum_i (x_i - \bar{x})y_i}{\sum_i (x_i - \bar{x})^2}|X\right) \\
&= \frac{\mathrm{Var}\left(\sum_i (x_i - \bar{x})y_i|X\right)}{\left(\sum_i (x_i - \bar{x})^2\right)^2} \\
&= \frac{\sum_i (x_i - \bar{x})^2 \sigma^2}{\left(\sum_i (x_i - \bar{x})^2\right)^2} \\
&= \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}.
\end{aligned}$$

Alternatively

$$\mathrm{Var}(\hat{\beta}_1|X) = \mathrm{Var}(\beta_1 + \sum w_i \epsilon_i|X) = \sum w_i^2 \mathrm{Var}(\epsilon_i|X) = \sum w_i^2 \sigma^2 = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}.$$

The variance of $\hat{\beta}_0$ is:

$$\begin{aligned}\mathrm{Var}(\hat{\beta}_0) &= \mathrm{Var}(\bar{y} - \hat{\beta}_1 \bar{x}) \\ &= \mathrm{Var}(\bar{y}) + \bar{x}^2 \mathrm{Var}(\hat{\beta}_1) \\ &= \frac{\sigma^2}{n} + \bar{x}^2 \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}.\end{aligned}$$

This properties, especially the sample distribution, allows to perform inference on β_1. Namely, testing hypothesis, constructing confidence intervals, etc.

Normality of OLS Estimators

By assuming that $\epsilon_i \sim^{iid} N(0, \sigma^2)$, since the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are weighted sum of the error terms, they also have normal distribution such that

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2_{\beta_j}),$$

where:

$$\sigma^2_{\beta_1} = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}, \quad \sigma^2_{\beta_0} = \frac{\sigma^2}{n} + \bar{x}^2 \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}.$$

Definition: Consistency
An estimator $\hat{\beta}_j$ (for $j = 0, 1$) is consistent if:

$$\lim_{n \to \infty} \mathbb{P}(|\hat{\beta}_j - \beta_j| > \epsilon) = 0, \quad \forall \epsilon > 0.$$

This means that $\hat{\beta}_j$ converges in probability to β_j as the sample size $n \to \infty$.
The weak law of large numbers

- Let $X_1, ..., X_n$ be i.i.d random variables, thus for every $\epsilon < 0$,

$$\lim_{n \to \infty} \mathbb{P}(|\bar{X}_n - \mathbb{E}[X]) > \epsilon) = 0,$$

  we denote it by $\bar{X}_n \xrightarrow{p} \mathbb{E}[X]$ as $n \to \infty$.

Heuristic proof (informal):

$$\frac{1}{n}\sum(x_i-\bar{x})(y_i-\bar{y}) \xrightarrow{p} \mathbb{E}[(X-\mathbb{E}X)(Y-\mathbb{E}Y)] = \text{cov}(X,Y), \text{as } n\to\infty$$

$$\frac{1}{n}\sum(x_i-\bar{x})^2 \xrightarrow{p} \mathbb{E}[(X-\mathbb{E}X)^2] = \text{Var}(X), \text{as } n\to\infty$$

Them, by the continuous mapping theorem,

$$\begin{aligned}
\hat{\beta}_1 = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2} &\xrightarrow{p} \frac{\text{cov}(X,Y)}{\text{Var}(X)} \\
&= \frac{cov(X,\beta_0+\beta_1X+\epsilon)}{Var(X)} \\
&= \frac{\beta_1 cov(X,X)}{Var(X)} \\
&= \beta_1\frac{Var(X)}{Var(X)} \\
&= \beta_1.
\end{aligned}$$

Alternatively, as $n \to \infty$

$$\begin{aligned}
\hat{\beta}_1 &= \beta_1 + \frac{\frac{1}{n}\sum(x_i - \bar{x})\epsilon_i}{\frac{1}{n}\sum(x_i - \bar{x})^2} \\
&\xrightarrow{p} \beta_1 + \frac{\mathbb{E}[(X - \mathbb{E}[X])\epsilon]}{\mathrm{Var}(X)} \\
&= \beta_1,
\end{aligned}$$

since

$$\mathbb{E}[(X - \mathbb{E}[X])\epsilon] = \mathbb{E}[\mathbb{E}[(X - \mathbb{E}[X])\epsilon|X]] = \mathbb{E}[(X - \mathbb{E}[X])\mathbb{E}[\epsilon|X]] = \mathbb{E}[(X - \mathbb{E}[X])0] = 0.$$

- For the causal model is

$$Y_i = \psi_0 + \psi_1 X_i + f(U_i) + u_i$$

The OLS estimator of the causal effect ψ_1 in the presence of unmeasured confounders is inconsistent.

- By using the weak low if large number (WLLN) and the continuous mapping theorem, we have

$$\psi_{\text{OLS}} = \frac{\widehat{\text{cov}}(X, Y)}{\widehat{\text{Var}}(X)} \xrightarrow{p} \frac{\text{cov}(X, Y)}{\text{Var}(X)},$$

where $\text{cov}(X, Y) = \text{cov}(X, \psi_0 + \psi_1 X + f(U) + \epsilon),$

$$\text{cov}(X, Y) = \psi_1 \text{cov}(X, X) + \text{cov}(X, f(U)),$$

$$= \psi_1 \text{Var}(X) + \text{cov}(X, f(U)).$$

- Therefore,

$$\psi_{\text{OLS}} \xrightarrow{p} \psi_1 \frac{Var(X)}{Var(X)} + \frac{\text{cov}(X, f(U))}{\text{Var}(X)} = \psi_1 + \frac{\text{cov}(X, f(U))}{\text{Var}(X)} \neq \psi_1.$$

Problem framework

- The structural equation is:

$$Y = \psi_0 + \psi_1 X + f(U) + u,$$

where:

  - $f(U)$ introduces endogeneity (confounding factor),
  - X is endogenous, and u is an error term.

- The instrument Z satisfies:
  - Relevance: $\text{Cov}(Z, X) \neq 0$,
  - Exogeneity: $\text{Cov}(Z, u) = 0$.

$$\begin{tikzcd} & & U \arrow[dl] \arrow[dr] & \\ Z \arrow[r] & X \arrow[rr, "\psi_1"] & & Y \end{tikzcd}$$

IV estimator: Two Steps Least Square (2SLS or TSLS)

- Assume that now we observe additionally an instrument Z, thus the observed data is $\{y_i, x_i, z_i\}_{i=1}^n$.
- We can estimate the causal effect of X on Y in two steps:
  1. **First stage**: Regress X on Z and obtain predicted values of $\hat{X}$, i.e., estimate
  $$X = \alpha_0 + \alpha_1 Z + \epsilon.$$
  2. If $\mathrm{Cov}(Z, u) = 0$, then $\hat{X}$ contains variation in X that is uncorrelated with u.
  3. Use the predicted values of X, i.e.,
  $$\hat{X} = \hat{\alpha}_0 + \hat{\alpha}_1 Z$$
  4. **Second stage**: Regress Y on $\hat{X}$, i.e.,
  $$Y = \psi_0 + \psi_1 \hat{X} + \epsilon_Y,$$
  5. Obtain the Two Stage Least Squares estimator $\hat{\psi}_{1,2SLS}$.
  $$\hat{\psi}_{1,2SLS} = \frac{\sum(Y_i - \bar{Y})(\hat{X}_i - \bar{\hat{X}})}{\sum(\hat{X}_i - \bar{\hat{X}})^2}$$

Definition

The ratio of the covariance between Z and Y to the covariance between Z and X:

$$\frac{\mathrm{cov}(Z, Y)}{\mathrm{cov}(Z, X)},$$

is known as the Wald ratio, which is the ATE ψ_1 in the liner causal model $Y = \psi_0 + \psi_1 X + f(U) + u$, as

$$cov(Z, Y) = cov(Z, \psi_0 + \psi_1 X + f(U) + u) = cov(Z, \psi_1 X) = \psi_1 cov(Z, X)$$

thus,

$$\frac{\mathrm{cov}(Z, Y)}{\mathrm{cov}(Z, X)} = \psi_1 \frac{cov(Z, X)}{cov(Z, X)} = \psi_1.$$

Special Case for Binary Variables

For binary Y and X, this outcome is the ITT (Intention-to-Treat) to compliance ratio, i.e.,

$$\frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[X|Z=1] - \mathbb{E}[X|Z=0]} = \frac{\text{ITT}}{\text{Compliance ratio}}.$$

The 2SLS estimator is equal to the Wald sample ratio

$$
\begin{aligned}
\hat{\psi}_{1,2SLS} &= \frac{\sum(Y_i - \bar{Y})(\hat{X}_i - \bar{\hat{X}})}{\sum(\hat{\alpha}_0 + \hat{\alpha}_1 X_i - \bar{X})^2} \\
&= \frac{\sum(Y_i - \bar{Y})(\hat{\alpha}_0 + \hat{\alpha}_1 Z_i - \bar{X})}{\sum(\hat{\alpha}_0 + \hat{\alpha}_1 Z_i - \bar{X})^2} \\
&= \frac{\sum(Y_i - \bar{Y})(\bar{X} - \hat{\alpha}_1 \bar{Z} + \hat{\alpha}_1 Z_i - \bar{X})}{\sum(\bar{X} - \hat{\alpha}_1 \bar{Z} + \hat{\alpha}_1 Z_i - \bar{X})^2} \\
&= \frac{\sum(Y_i - \bar{Y})(Z_i - \bar{Z})}{\hat{\alpha}_1 \sum(Z_i - \bar{Z})^2} \\
&= \frac{\sum(Y_i - \bar{Y})(Z_i - \bar{Z})}{\sum(Z_i - \bar{Z})^2} \times \frac{\sum(Z_i - \bar{Z})^2}{\sum(X_i - \bar{X})(Z_i - \bar{Z})} \\
&= \frac{\sum(Y_i - \bar{Y})(Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})} \\
&= \frac{\widehat{cov}(Z, Y)}{\widehat{cov}(Z, X)} \quad \square
\end{aligned}
$$

- The ratio of the sample covariance between Z and Y to the sample covariance between Z and X:
$$\frac{\widehat{\text{cov}}(Z,Y)}{\widehat{\text{cov}}(Z,X)}$$
converges to the true covariance ratio: $\frac{\text{cov}(Z,Y)}{\text{cov}(Z,X)}$, which is the ATE ψ_1.
- Alternatively, we can write the 2SLS estimator in the same fashion as the OLS estimator, i.e.,
$$\begin{aligned}
\hat{\psi}_{2SLS} &= \frac{\sum(Y_i - \bar{Y})(Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})} \\
&= \frac{\sum\left[(\psi_0 + \psi_1 X_i + f(U_i) + u_i) - (\psi_0 + \psi_1 \bar{X} + \bar{f}(U_i) + \bar{u})\right](Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})} \\
&= \frac{\sum \psi_1 (X_i - \bar{X})(Z_i - \bar{Z}) + \sum(f(U_i) + u_i - \bar{f}(U) - \bar{u})(Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})},
\end{aligned}$$
define $f(U_i) + u_i = \tilde{u}_i$, thus
$$\hat{\psi}_{2SLS} = \psi_1 + \frac{\sum(\tilde{u}_i - \bar{\tilde{u}})(Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})} = \psi_1 + \frac{\sum \tilde{u}_i (Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})}.$$

2SLS Consistency

- Convergence in probability:

$$\begin{aligned}\hat{\psi}_{2SLS} &= \psi_1 + \frac{\sum \tilde{u}_i(Z_i - \bar{Z})}{\sum (X_i - \bar{X})(Z_i - \bar{Z})} \\ &\xrightarrow{p} \psi_1 + \frac{cov(\tilde{u}, Z)}{cov(X, Z)}\end{aligned}$$

- Where

$$cov(\tilde{u}, Z) = cov(f(U) + u, Z) = cov(f(U), Z) + cov(u, Z),$$

since Z is exogenious, for $n \to \infty$

$$cov(f(U), Z) = cov(u, Z) = 0.$$

- Thus, $\hat{\psi}_{2SLS}$ is consistent estimator of ψ_1.

2SLS Bias Illustration

- The expectation of the 2SLS estimator is:

$$\mathbb{E}[\hat{\psi}_{1,2SLS}] = \psi_1 + \mathbb{E}\left[\frac{\sum \tilde{u}_i(Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})}\right].$$

- Where

$$\mathbb{E}\left[\frac{\sum \tilde{u}_i(Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})}\right] = \mathbb{E}\left[\frac{\sum \mathbb{E}[\tilde{u}_i|X_i, Z_i](Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})}\right]$$

- Notably, instrument exogeneity implies $cov(Z, f(U)) = 0$ which implies consistency of the 2SLSL, but $\mathbb{E}[f(U)|Z, X] \neq 0$, then, generally the 2SLS is biased for every finite sample size.
- The bias arises because $\text{Cov}(\hat{X}, f(U)) \neq 0$ when $\hat{X}$ is correlated with the confounder $f(U)$.

2SLS Sample variance

- The expectation of the 2SLS estimator is:

$$Var[\hat{\psi}_{1,2SLS}] = \psi_1 + Var\left[\frac{\sum \tilde{u}_i(Z_i - \bar{Z})}{\sum(X_i - \bar{X})(Z_i - \bar{Z})}\right].$$

- The exact form depends on

$$Var((Z - \mathbb{E}[Z])(u + f(U)) = Var((Z - \mathbb{E}[Z])f(U))$$

- The 2SLS estimator is **consistent**:

$$\hat{\psi}_{1,2SLS} \xrightarrow{p} \psi_1 \quad \text{as} \quad n \to \infty.$$

- It is **asymptotically normal**:

$$\sqrt{n}(\hat{\psi}_{1,2SLS} - \psi_1) \xrightarrow{d} \mathcal{N}(0, \sigma^2),$$

where σ^2 is the asymptotic variance:

$$\sigma^2 = \frac{\text{Var}((Z - \mathbb{E}[Z])\tilde{u})}{\text{Cov}^2(Z, X)}.$$

- We will revise and elaborate on the variance calculations in the next lectures.

Finite-Sample Considerations

- In **small samples**, the distribution of $\hat{\psi}_{1,2SLS}$ may deviate from normality due to:
  - Small sample size (n),
  - Weak instruments ($\text{Cov}(Z, X)$ is small).
- The finite-sample bias arises because:

$$\text{Cov}(\hat{X}, f(U)) \neq 0,$$

  where $\hat{X}$ is the predicted value of X from the first-stage regression. This bias is amplified when Z is weak or correlated with $f(U)$.
- Inference in finite samples may require:
  - **Robust methods**, such as bootstrap procedures, to account for non-normality.
  - **Weak-instrument robust tests**, such as the Anderson-Rubin test, which do not rely on strong instrument assumptions.
- Nevertheless, the normal approximation of the finite sample distribution of the 2SLS estimator allows for computation os the analytical confidence intervals (CIs) and hypothesis testing.
- If the assumed models are correctly specified, these CIs tend to be narrower than the bootstrap-based CIs.

Definition:
Weak instruments are instrumental variables (IVs) that have a weak or insufficient correlation with the endogenous explanatory variable (X) in a regression model. Formally, an instrument Z is considered **weak** if its relevance condition ($\text{Cov}(X, Z) \neq 0$) is satisfied but the strength of the relationship between Z and X is very small.

Characteristics of Weak Instruments:

- Low correlation between X and Z:
  - The instrument Z explains only a small portion of the variation in X.
- Weak first-stage regression:
  - In the first-stage regression of X on Z, the F-statistic for the instrument Z is low ($F < 10$ is often considered problematic).

Problems Caused by Weak Instruments

$$\hat{\psi}_{1,2SLS} = \psi_1 + \frac{\sum_{i=1}^{n}(Z_i - \bar{Z})u_i}{\sum_{i=1}^{n}(Z_i - \bar{Z})(X_i - \bar{X})},$$

- Biased Estimates:
  - Weak instruments lead to biased estimates of the causal effect ψ in finite samples, even when other IV assumptions hold.
- Large Standard Errors:
  - The standard errors of the estimated coefficients become inflated, reducing statistical power.
- Inconsistency:
  - As the strength of the instrument diminishes, the two-stage least squares (2SLS) estimator can converge to a value far from the true causal effect, even in large samples.

Global F-test

For the general linear regression model:

$$y_i = \beta_0 + \sum_{j=1}^{p} \beta_j X_{j,i} + \epsilon_i$$

- The tested hypothesis is $H_0 : \beta_1 = ... = \beta_p = 0$, namely, none of the predictors significantly contributes to explaining the variability of Y.
- The test statistic is

$$F = \frac{MSreg}{MSE},$$

where MSReg is the $SSreg/p$ and MSE is the unbiased estimator of $\sigma^2 = Var(\epsilon|X)$.
- In normal linear models, under H_0, the test-statistic F is distributed $F(p, n-p-1)$.
- Therefore, large values of F indicate that at least one of the predictors is useful in predicting/explaining y.

Partial F-test

- The tested hypotheis is $H_0 : \beta_1 = ... = \beta_q = 0$, $q < p$, namely q out of the p predictors have no contribution in explaining y.
- We can reformulate the hypotheses as $H_0 : RM$ vs. $H_1 : FM$, where RM is the reduced (or restricted) model, and FM is the full model.
- The test statistic is

$$F = \frac{(SSreg(FM) - SSreg(RM))/(df(FM) - df(RM))}{MSE(FM)} = \frac{\Delta SSreg/q}{MSE},$$

  where $\Delta SSreg$ is the difference in the explained sum of squares between the models, q is the number of restricted coefficients, and MSE is the MSE of the full model.
- Under H_0, the test-statistic F is distributed $F(q, n - p - 1)$.
- Note that the models have to be:
  (1) linear with a normally distributed error term
  (2) nested, i.e., RM should be nested in the FM.

Detecting and addressing weak instruments:

Detecting Weak Instruments:

- F-Statistic in the First Stage:
  - In the first-stage regression of X on Z, a rule of thumb is that the F-statistic for the instrument should be greater than 10.
  - If $F < 10$, the instrument is considered weak.

Addressing Weak Instruments:

- Use Stronger Instruments:
  - Find additional or alternative instruments with a stronger relationship to X.
- Use LIML (Limited Information Maximum Likelihood):
  - LIML is more robust to weak instruments compared to 2SLS.
- Robust Inference Methods:
  - Tests like the Anderson-Rubin test can provide valid inference even with weak instruments.

Classification of the identification conditions

- When there are multiple treatment variables the 2SLS methods is similar except that each treatment variable requires its own first stage model.
- For IV regression to be possible there should be at least as many instruments as treatment variables.
- Let M be the number of instruments and K the number of treatment variables: The model is said to be:

    1. Underidentified if $M < K$, we cannot estimate the model, the number of instruments is then smaller that the number of treatment variables.
    2. Exactly identified if $M = K$, the number of instruments equals the number of treatment variables.
    3. Overidentified if $M > K$, the number of instruments exceeds the number of treatment variables.

- If there are more instruments M than treatment variables, then there is a statistical test that test (partially) for exogeneity of the instruments.
- In such case, we could calculate M different IV-estimates of ψ_1.
- Since any linear combination of the Z_{mi} is again a valid instrument:
  - Combine the Z_{mi} to get a more efficient estimator of ψ_1.

$$X_i = \alpha_0 + \alpha_1 Z_{1i} + \cdots + \alpha_M Z_{Mi} + \gamma_1 L_{1i} + \cdots + \gamma_r L_{ri} + \epsilon_{X,i}.$$

Instrumental Variable Assumptions:

1. Instrument relevance:
   - At least one of the instruments $Z_{1i}, \ldots, Z_{Mi}$ should have a nonzero coefficient in the population regression of X_i on $Z_{1i}, \ldots, Z_{Mi}$.
2. Instrument exogeneity:
   - $\text{Cov}(Z_{1i}, u_i) = \text{Cov}(Z_{2i}, u_i) = \cdots = \text{Cov}(Z_{Mi}, u_i) = 0$.

- Suppose we have two instruments Z_1, Z_2. Given the structural equation:

$$Y_i = \psi_0 + \psi_1 X_i + \beta_1 L_{1i} + \cdots + \beta_r L_{ri} + u_i,$$

  and assuming that $\text{Cov}(Z_{1i}, u_i) = 0$. Namely, the first instrument Z_1 is valid, then we can test whether $\text{Cov}(Z_{2i}, u_i) = 0$, i.e., the second instrument Z_2 is also valid.
- Analogically, we can assume $\text{Cov}(Z_{2i}, u_i) = 0$, and then test whether $\text{Cov}(Z_{1i}, u_i) = 0$.
- Testing whether $\text{Cov}(Z_{2i}, u_i) = 0$ or $\text{Cov}(Z_{1i}, u_i) = 0$ is equivalent to testing:

$$\hat{\psi}_1^{(Z_2)} = \hat{\psi}_1^{(Z_1)}.$$

Test implementation:

1. Estimate the model ($L_1, \ldots, L_r$ are measured confounders):

$$Y_i = \psi_0 + \psi_1 X_i + \beta_1 L_{1i} + \cdots + \beta_r L_{ri} + u_i,$$

   by 2SLS using Z_{1i} and Z_{2i} as instruments.

2. Obtain the residuals:

$$\hat{u}_i^{2SLS} = Y_i - \hat{\psi}_0 - \hat{\psi}_1 X_i - \hat{\beta}_1 L_{1i} - \cdots - \hat{\beta}_r L_{ri}.$$

3. Estimate the following regression model:

$$\hat{u}_i^{2SLS} = \theta_0 + \theta_1 Z_{1i} + \theta_2 Z_{2i} + \delta_1 L_{1i} + \cdots + \delta_r L_{ri} + e_i.$$

4. Obtain the partial F-statistic of the following test:

$$H_0 : \theta_1 = \theta_2 = 0.$$

5. Compute the J-test statistic:

$$J = mF \sim \chi_q^2,$$

   where m is the number of instruments, and q is the number of instruments minus the number of endogenous regressors (i.e., the treatment variables).

Overidentifying Restrictions Test - J test

Although the J-test is a useful test, there are 3 main reasons to be careful when using it:

1. When we don't reject the null hypothesis, this does not mean that we can accept it, but rather that we don't have enough "evidence" to reject it.
2. The power of the J-test can be low.
3. The J-test tests the joint hypothesis of instrument validity and correct functional form: If we reject the null hypothesis, the instruments might be valid but the functional form is wrong.

References

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